# LIFTING FINITE AND FINITELY PRESENTED MODULES ACROSS A LOCALISED QUOTIENT

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*Allowed standard prerequisites (editorial).*

- Preimages of submodules under a surjection. Used in part (1) of the counted proof (lines 31722-31725). Produces the finite module over R whose localised quotient is the given module.
- Clearing denominators in a matrix over a localised quotient. Used in part (2) of the counted proof (lines 31732-31735). Replaces the presentation matrix over R' by a matrix over R after multiplying by an element of the multiplicative subset.

**Lemma 0.1.** *Let $R$ be a ring. Let $I \subset R$ be an ideal. Let $S \subset R$ be a multiplicative subset. Set $R' = S^{-1}(R/I) = S^{-1}R/S^{-1}I$.*

(1) *For any finite $R'$-module $M'$ there exists a finite $R$-module $M$ such that $S^{-1}(M/IM) \cong M'$.*
(2) *For any finitely presented $R'$-module $M'$ there exists a finitely presented $R$-module $M$ such that $S^{-1}(M/IM) \cong M'$.*

*Proof.* Proof of (1). Choose a short exact sequence $0 \to K' \to (R')^{\oplus n} \to M' \to 0$. Let $K \subset R^{\oplus n}$ be the inverse image of $K'$ under the map $R^{\oplus n} \to (R')^{\oplus n}$. Then $M = R^{\oplus n}/K$ works.

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Proof of (2). Choose a presentation $(R')^{\oplus m} \to (R')^{\oplus n} \to M' \to 0$. Suppose that the first map is given by the matrix $A' = (a'_{ij})$ and the second map is determined by generators $x'_i \in M'$, $i = 1, \ldots, n$. As $R' = S^{-1}(R/I)$ we can choose $s \in S$ and a matrix $A = (a_{ij})$ with coefficients in $R$ such that $a'_{ij} = a_{ij}/s \bmod S^{-1}I$. Let $M$ be the finitely presented $R$-module with presentation $R^{\oplus m} \to R^{\oplus n} \to M \to 0$ where the first map is given by the matrix $A$ and the second map is determined by generators $x_i \in M$, $i = 1, \ldots, n$. Then the map $M \to M'$, $x_i \mapsto x'_i$ induces an isomorphism $S^{-1}(M/IM) \cong M'$. $\square$